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TO:

All OSD Inspection Team Leaders

# CONTROL SYSTEMS FOR ESD

## PURPOSE

This SPC describes the control system aspects of Emergency Shutdown (ESD) Systems in offshore oil and gas production systems and establishes a baseline as to what may be regarded as 'appropriate measures' required by PFEER. This SPC should be read in conjunction with SPC/Tech/OSD/09, where certain basic control system matters relevant to ESD systems are discussed. However, design features and operational matters specific to ESD systems, and their management, are considered in this paper. Matters appropriate for a general inspection or investigation are specifically identified; discussions at a deeper level are intended to provide a baseline to underpin a consistent approach across the industry.

This SPC builds on and updates Section 91 of the Fourth Edition Guidance and other industry guidance, and seeks to interpret the requirements of PFEER in respect of ESD systems in the offshore oil and gas industry. Certain material common to all control systems is discussed in Appendix A of SPC/Tech/OSD/09, which should be read in conjunction with this SPC.

## ACTION

Inspection teams should consider the use of the question sets in Section 8 for inspection work or for investigations of incidents. For each question, a model answer is given to indicate typical best practice. Where a duty holder uses a different approach to that indicated in the model answer, but achieves a similar level of management control, this is satisfactory; however, if management controls are absent or of low quality, further consideration is recommended.

## BACKGROUND

## 1 THE ROLE OF ESD SYSTEMS

1.1 ESD system protect against the possibility of a process excursion on topsides process plant developing into an incident (eg loss of containment), and to respond to emergency situations detected by other safeguarding facilities. This protection is part of a hierarchy provided by a number of layers, typically:

(i) Process control, including operator intervention via alarm functions;

(ii) Instrument trip and ESD functions; and

(iii) Self acting mechanical protection such as relief valves or bursting discs.

1.2 Some of the shut down functionality in (i) and (ii) above may be called 'process shut down' rather than ESD. Note that protection against a process excursion on a different installation connected by a pipeline is not covered by this SPC.

1.3 Modulating control systems are designed to contain normal process deviations such as those caused by irregular slug flow; in addition, operator intervention (for example following alarms generated by the process control system) is intended to maintain safe and effective production by reducing the extent of process upsets and hence the demand rate on the formal safeguards. However, it is difficult to take formal credit for process control and alarms (see SPC/Tech/OSD/09), so ESD and related instrumented trip functions are regarded as the 'primary' automatic safeguarding function. Mechanical devices are regarded as 'secondary' protection in the sense that they act after the 'primary' safeguard, though mechanical devices provide a fundamental or 'ultimate' safeguard.

1.4 Where mechanical protection is provided, the result of a failure to act by an instrumented trip places a demand on the mechanical protection and there should be no safety consequences if the mechanical protection is correctly designed. There may be environmental consequences caused eg by the escape of product or asset loss consequences caused by the need to refurbish the mechanical protective device. In some designs, the quantitative safety improvement required of the protective function is formally partitioned between the ESD trip function and the mechanical protection. Where the instrumented protection is the only automatic protection and independent mechanical protection is not provided, safety hazards arise if the ESD function fails to act in response to a demand; such ESD functions need to be of higher integrity in order to meet the entire performance requirement; such applications were formerly known as 'HIPS' (High Integrity Protective Systems), see companion SPC/Tech/OSD/09 Appendix A.

1.5 The ESD system works by interpreting a number input signals from plant measurements, and executing a 'cause and effect' logic to shut down, isolate or vent predetermined items of plant, or to initiate safety systems, according to the nature, location and severity of any hazard. Manual inputs to the system are provided so that installation personnel may shut down the plant in response to conditions not covered by automatic protection (perhaps for example a minor leak which might ultimately develop into a safety, environment or asset protection problem).

1.6 The other major function of the ESD system is to execute inter-trips from other systems, most obviously the fire & gas detection system. The inter-trip function implements similar 'cause and effect' logic; for example, in the case of a fire and gas system inter-trip this is designed (typically) to vent and partition the topsides inventory into smaller volumes to limit the effects of a loss of containment feeding a fire. Also, the ESD system may pass demands to other systems, eg the GPA system, HVAC controls, electrical isolations, etc.

1.7 The ESD system should be designed to implement the process safety intent, and it should be designed to perform with adequate availability and survivability. It is equally important for it to be operated, maintained and modified in such a way as to continue to meet the design safety intent whilst in service.

1.8 ESD systems are dormant in normal service and should therefore be designed so that failures are self-revealing or detected by built in test. Proof testing should also be carried out (as described in Section 7), with a particular focus on components not covered by built-in test.

## 2 LEGAL REQUIREMENT IN PFEER

2.1 Certain specific legal requirements for process and utility control systems are to be found in PFEER, usually expressed in terms of ‘appropriate measures’. It is important to be aware of the specific meaning of the word ‘emergency’ which is used in Regulations 10-12; it is defined as ‘an emergency of a kind which can require evacuation, escape or rescue'. It is likely that only large-scale incidents could be so described. The term ‘major accident’ used in Regulation 5 is defined by the Safety Case Regulations, and in the context of this paper the definition reduces to ‘a fire, explosion or the release of a dangerous substance involving death or serious personal injury to persons on the installation....’. Regulation 9 has no caveats as to the size of incident.

Regulation 5 requires an assessment including the:

- ‘establishment of appropriate standards of performance to be attained by anything provided by measures for:

(ii) otherwise protecting persons from a major accident involving fire and explosion'

Regulation 9 requires 'appropriate measures with a view to preventing fire and explosion, including such measures to':

- 'ensure the safe production, processing, ... and other dealings with flammable ... substances'
- 'prevent the uncontrolled release of flammable or explosive substances'
- 'prevent the unwanted or unnecessary accumulation of combustible flammable or explosive substances'

Regulation 10 requires 'appropriate measures with a view to'

- 'detecting ... events which may require an emergency response'
- 'enabling information regarding such incidents to be conveyed forthwith to a place from which control action can be instigated'

Regulation 11 requires 'appropriate arrangements'

- 'for giving warning of an emergency'

Regulation 12 requires

- 'appropriate measures with a view to limiting the extent of an emergency, including fire and explosion';
- 'those measures (shall) include provision for the remote operation of the plant';
- 'so far as is reasonably practicable, arrangements and plant provided pursuant to this Regulation (shall be) capable of remaining effective in an emergency';

Regulation 13 requires

- 'appropriate measures with a view to protecting persons on the installation during an emergency from the effects of fire and explosion'.

2.2 The PFEER ACoP interprets the basic requirements, as follows:

- Communication 'arrangements' should be based on the findings of the assessment required by Regulation 5 for major hazards, and on Regulation 3 of MHSWR for non-major hazards.
- Measures against 'major accidents' to be based on the assessment required by Regulation 5.

- Emergency shutdown should be capable of initiation from the control point.

2.3 Thus Regulations 9-13 outline the FUNCTIONS required; Regulation 5 requires the PERFORMANCE required of those functions to be defined. On a typical installation, many of these functions are provided (at least in part) by the ESD system itself, or involve the ESD system in responding to, or in initiating, inter-trips, as discussed in Section 3, Performance issues are discussed in Section 4.

2.4 The dividing line between appropriate measures and a less satisfactory arrangement which might be worthy of enforcement action is not defined in PFEER or in case law, but certain matters are identified below as being of specific concern. Also, any general shortfall in the standard of good practice defined in this SPC would be a cause for concern.

## 3 FUNCTION - CAUSE AND EFFECT REQUIREMENTS

3.1 ESD function logic is generated from the overall process design and safety studies, and traditionally is expressed in a matrix which relates ‘causes’ (eg sensor inputs) to ‘effects’ (eg valve closures). These 'cause and effect diagrams' specify (though not necessarily with truly logical completeness) the functional requirements of the ESD system.

3.2 It is normal to define several levels of shut down related to the nature of the hazard. An event (cause) on an individual plant item, with little or no potential to escalate and affect other plant areas, may attract the lowest level of 'unit shut down' (effect). Depending on the complexity of the plant and the location the nature of the cause, more widespread shut downs come into play, to give a hierarchy of shut-downs. A typical structure might be as follows:

- unit shut down
- train shut down
- production shutdown (with no blow down)
- production shutdown (with blow down)
- platform shutdown (with power generation shut down)
- abandon platform shutdown (with complete electrical isolation)

3.3 The lower levels of this hierarchy are often implemented as ‘process shutdowns’ in separate (lower integrity) hardware from the higher level ‘emergency shutdowns’. However, there is no absolute connection between shut down level and the required performance. A ‘low level’ unit shutdown may require high reliability if it protects against a severe hazard. The higher levels of shut down may not require extraordinary performance since they are called upon very rarely. Each individual function should be considered on its own merits (see Section 4).

3.4 Some of the implied functionality is based on ‘inter-trips’. Typical inter-trips originate in the fire and gas system, and instruct the ESD system to execute emergency isolation and venting (as defined by the cause and effect charts) on confirmed fire or confirmed gas release.

## 4 PERFORMANCE - SAFETY INTEGRITY LEVELS

4.1 Process protection facilities should have a defined function, typically expressed in the cause and effect charts (eg to shut a given valve at a given pressure), and defined survivability and availability. The availability is often specified in terms of a Safety Integrity Level (SIL). PFEER Regulation 5 implies that any credible hazard that could produce death or injury from fire or explosion should be formally managed; hence related protective functions should be of SIL 1 or higher, as described in ref 1.

4.2 Survivability is not often an issue because most faults cause a failure to a safe state (see Section 5), thus arguably, the SIL of a given function largely expresses its ‘performance standard’ in the sense of PFEER.

## 5 ESD SYSTEM HARDWARE

5.1 ESD functions are implemented by means of 3 basic elements: process sensors, computation, and plant final control, as discussed in some detail in SPC/Tech/OSD/09.

### Sensors

5.2 Typical ESD sensors are pressure switches or level switches, but analogue sensors or transmitters can also used to generate ON/OFF signals by means of a trip amplifier (which produces a switched output at a pre-set trip setting); indeed, analogue sensors generally give higher reliability as they are continuously exercised, whereas switches are dormant and may fail to danger. The use of a DCS or similar may allow the outputs of redundant analogue sensors to be compared, so that fault conditions can be detected before they become significant.

### Final Control Elements

5.3 In ESD functions, typical final control elements are shut-off valves, vent valves, motor start/stop, etc. Higher SIL function normally require two ESD valves in series.

### Computation

5.4 ESD computation is in logic form, eg AND and OR functions, on process plant signals to provide appropriate interlocks and control signals to implement the cause and effect requirements. PLCs are commonly used for ESD functions at the lower integrity levels. The use of PLCs at higher integrity levels would be harder to justify, and ref 1 advises against the use of software based systems for SIL 3 applications unless particularly rigorous procedures are

followed. It is usual to segregate high criticality safety functions into their own specifically designated non-programmable safety system.

## 6 MINIMUM PROVISION OF ESD FUNCTIONS

6.1 The minimum provision of ESD functions that should be provided on a typical offshore oil and gas plant is as follows (derived from ref 2). Any shortfall against this guidance should be viewed seriously.

6.2 In general, each section of a pressure system with a credible connection to a source of pressure exceeding its rating (typically those sections protected by process relief valves or equivalent, though not fire relief valves) should be provided with a high pressure trip. Large sections of the system that can be isolated should have their own trips, eg where there is more than one stream, each stream should have its own trips. Given the inherent uncertainty of well shut-in pressures, it is prudent to provide high-pressure trips on wells and flowlines even if fully rated for the maximum expected shut-in pressure.

6.3 In certain cases it may be prudent to provide a low pressure trip, for example to shutdown in the event of a rupture of the pressure containment (eg of a flowline).

6.4 Process vessels (either pressure vessels or atmospheric tanks) which contain liquid levels should have high and/or low trips on the level in order to prevent possible liquid carry over, gas blowby, or contamination (eg of the water stream by oil) where these events have safety implications. Liquid carry over to compressors (especially reciprocating machines) is a common major safety concern, as is gas blowby to plant not designed to cope with the associated pressure increase.

6.5 Compressors should have high and low pressure trips on the suction and discharge lines, and a high temperature trip on the discharge line, plus non-process trips related to bearing temperature, vibration, etc, as required.

6.6 Fired vessels should have high and low level and temperature trips, plus trips related to the combustion process (typically high and low fuel pressure, low air pressure, and flame failure). Waste heat exchangers require only a high temperature trip.

6.7 Pumps should have high and low discharge pressure trips, plus non-process trips related to bearing temperature, vibration, etc, as required. Glycolpowered glycol pumps require low pressure trips on both inlet and discharge sides.

6.8 Shell-tube heat exchangers require high and low pressure trips on the process fluid inlet line and heating medium outlet line. A high temperature trip is not needed if both sections are fully rated for the maximum temperature of the heating medium.

6.9 Fire & gas events should trigger (via inter-trips) shut down of equipment in the relevant **area** of the installation. It is also necessary to shut down **systems** that may impact the hazard.

## 7 ESD SYSTEM OPERATION

7.1 Of fundamental importance to the successful operation and maintenance of an ESD system is the availability of properly controlled information on correct instrument trip settings and on system cause and effect requirements.

7.2 In order to achieve any given SIL over a period of service, it is necessary to test the function on a regular basis in order to identify otherwise unrevealed failures. The system design should incorporate a calculation of the required test frequency based on the system architecture and known component failure rates to achieve the intended SIL; this testing is known as 'trip testing' or 'proof testing'. Sensors can be calibrated and logic can be tested quite easily when an output override is applied to prevent any process action on the plant. Testing of final control elements such as shut down valves is more problematical since a process disturbance or shut down can result from the test, though partial valve movements can be a useful test. Testing of ESD system outputs is therefore usually carried out as part of planned plant shut downs, for example by simulating a demand on a given protective function. Testing can be carried out on an opportunistic basis when an actual demand or spurious shut down occurs, for example by scrutinising the event log to ensure that all 'effects' related to the 'cause' have been actioned within the prescribed time limit. Also, when plant items are shut down, it is possible to stroke **valves for test purposes.**

## 8 QUESTIONS AND ANSWERS

Typical 'Operational' Questions for Inspection or Investigative Questioning regarding 'Appropriate Measures' as required by PFEER

### A. General Questions and Model Answers

**Q: Are protective functions subject to periodic testing (trip testing)?**
A: Yes, there is a regular programme based on the criticality (SIL) of each trip. Most testing focuses on the process sensors and inputs to the electronic system (with the output inhibited), but the cause and effect logic is re-checked whenever any modification is carried out. Plant actioners can only be tested in a real trip (including spurious trips) or with the plant already shut down. This is done on an opportunistic basis, for example by checking the ESD system event printout for valve closure times.

**Q: Are inhibits controlled?**
A: Yes inhibits are co-ordinated by the permit controller and authorised by the Production Supervisor (as part of the permit process) when initially applied, and are re-checked by production personnel at every shift handover.

### B. Detailed Questions and Model Answers

**Q: Is operational experience of faults in ESD equipment and failures of functionality reconciled with assumptions made in the design?**
A: The responsible Instrument Engineer gathers information on faults found in the sensors, logic and actioners, and the failure rates and failure modes are compared with the design assumptions. Real demands on the system are monitored to establish the demand rate in service. When an ESD trip fails to operate, the cause is isolated and reconciled with the design SIL.

Typical 'Design' Questions for Inspection or Investigative Questioning regarding 'Appropriate Measures' as required by PFEER

## A. General Questions and Model Answers

**Q: Is a recognised methodology (eg ref 1) used to establish the required integrity of each protective function?**
A: Every specifically identified (and credible) hazard that could produce death or injury is formally managed and related protective functions are of SIL 1 or higher (ie protection is not totally reliant on operator responses to alarm functions). The UKOOA methodology (ref 1) is used for SIL assessment.

**Q: Does the DCS implement any safety functions, eg PSD?**
A. DCS is not used for safety functions of SIL 1 or above, and safety functions are segregated into their own specifically designated safety system.

## B. Detailed Questions and Model Answers

**Q: Have common mode failures been considered?**
A: Control system sensors and computation are independent of protection system components. Some final control elements are shared with process safeguarding actioners, as allowed by engineering standards (eg ref 1). In no case is a control element whose failure can cause a given process excursion employed in the protective function on the same or a related process variable. Alarm indications are, however, normally based on the **same** signals as used for control and may fail if, for example, the sensor fails.

## REFERENCES


1. ‘Guidelines for Instrument Based Protective Systems’, UKOOA, Rev2 1999.

2. ‘Recommended Practice for Analysis, Design, Installation and Testing of Basic Surface Safety Systems for Offshore Production Platforms’, API Recommended Practice 14C, 7th Edition, March 2001.


## FURTHER INFORMATION

Contact point for any questions: OD 3.5 Ext 8588